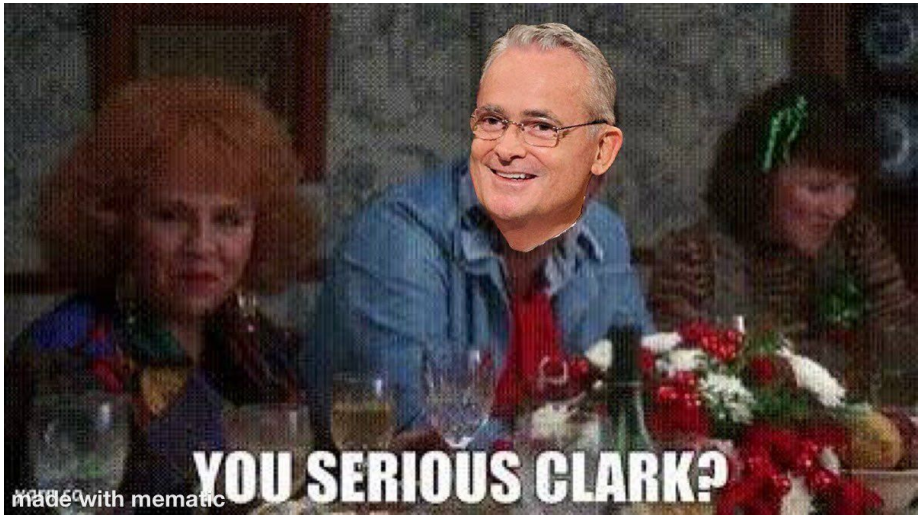
Perpendicular Alpha

Anders Neuenschwander and Brandon Waits

THE LAST DANCE

John after beating Peter to the tomb


YOU SERIOUS CLARK?
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Agenda

1. Thesis
2. Signal Construction
3. MVO Backtest Results

Changes Implemented

- Vol Scaling
- Gamma Tuning (now 5% active risk achieved)
- Comparison of industry momentum signal performance across standard momentum signal and idiosyncratic momentum

Signal Construction

1. Risk-Adjusted Factor Momentum and Asset-Level Projection

We calculate the volatility-scaled industry factor momentum using a 230-day lookback window with a 22-day lag. The raw momentum for each industry (the sum of daily factor returns) is divided by its exponentially weighted 22-day factor return volatility ($\sigma_{f,k,t}$) to ensure the signals are risk-adjusted. This yields a $K \times 1$ vector $\mathbf{m}_t$, which is then projected onto the asset space via the exposure matrix:

$$\underset{(K\times 1)}{\mathbf{m}_t} = \begin{bmatrix} \frac{\left(\sum_{\tau=t-230}^{t-22} f_{1,\tau}\right)}{\sigma_{f,1,t}} \\ \frac{\left(\sum_{\tau=t-230}^{t-22} f_{2,\tau}\right)}{\sigma_{f,2,t}} \\ \vdots \\ \frac{\left(\sum_{\tau=t-230}^{t-22} f_{K,\tau}\right)}{\sigma_{f,K,t}} \end{bmatrix} = \begin{bmatrix} m_{1,t} \\ m_{2,t} \\ \vdots \\ m_{K,t} \end{bmatrix}$$

$$\underset{(N\times 1)}{\mathbf{s}_t} = \underset{(N\times K)}{X_t} \times \underset{(K\times 1)}{\mathbf{m}_t}$$

Backtest Results

Performance Summary

Count	Mean Return (%)	Volatility (%)	Total Return (%)	Sharpe
3018	4.33	5.25	65.25	0.83

Drawdown Summary

Count	Mean Drawdown (%)	Max Drawdown (%)	Current Drawdown (%)	Longest Drawdown (days)
3018	-2.89	-10.07	-3.57	793

Leverage Summary

Count	Mean Leverage	Min Leverage	Max Leverage	Std Leverage
3018	359.31	255.55	455.54	53.16

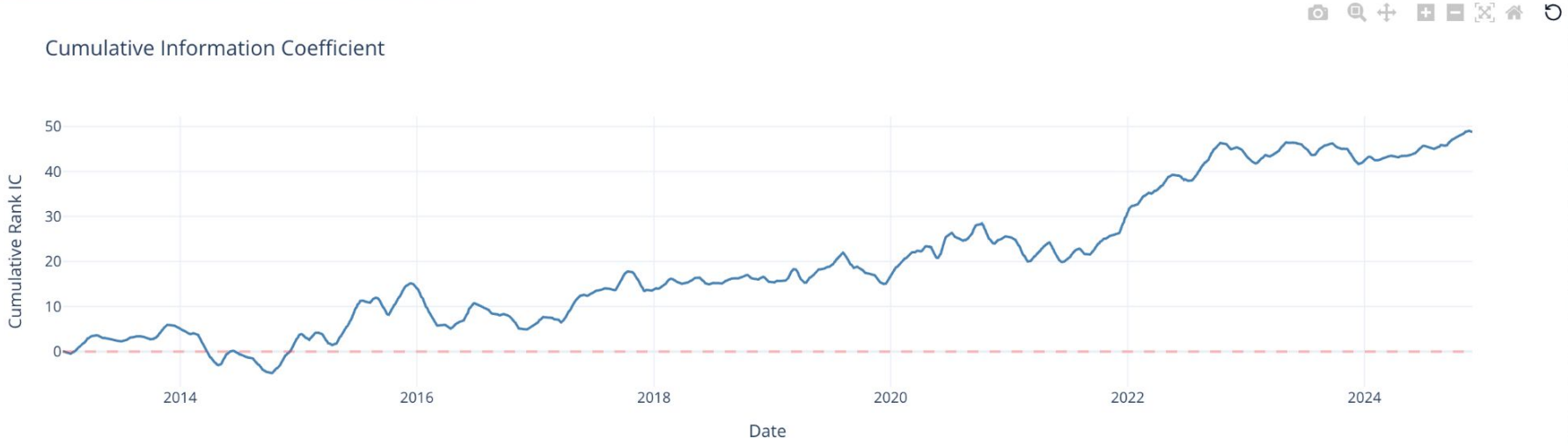
Cumulative Returns

Drawdown

Portfolio Drawdown
0
−2
−4
−6
−8
−10
Drawdown (%)
2014
2016
2018
2020
2022
2024
Date

Information Coefficient (IC)

Metric	Value
IC (mean)	**0.0163**
IC IR	**0.151**



Comparison

Correlation between Industry Momentum and Standard Momentum	Correlation between Industry Momentum and Idiosyncratic Momentum
0.25	0.13

Next Steps

VOTE!!!

Appendix

Standard Momentum

Date range: 2013-02-04 → 2024-12-31 **(2998 trading days)**

Performance Summary

Count	Mean Return (%)	Volatility (%)	Total Return (%)	Sharpe
2998	2.61	5	34.44	0.52

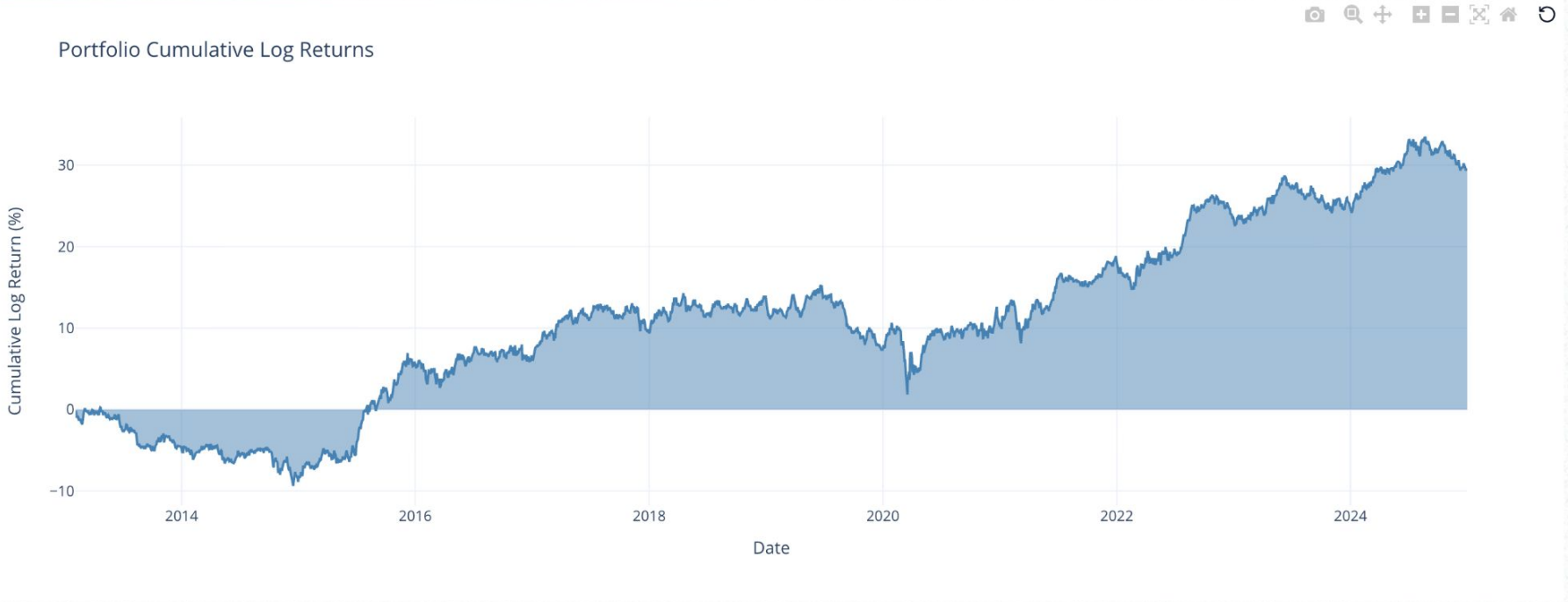
Drawdown Summary

Count	Mean Drawdown (%)	Max Drawdown (%)	Current Drawdown (%)	Longest Drawdown (days)
2998	-2.62	-12.6	-3.8	575

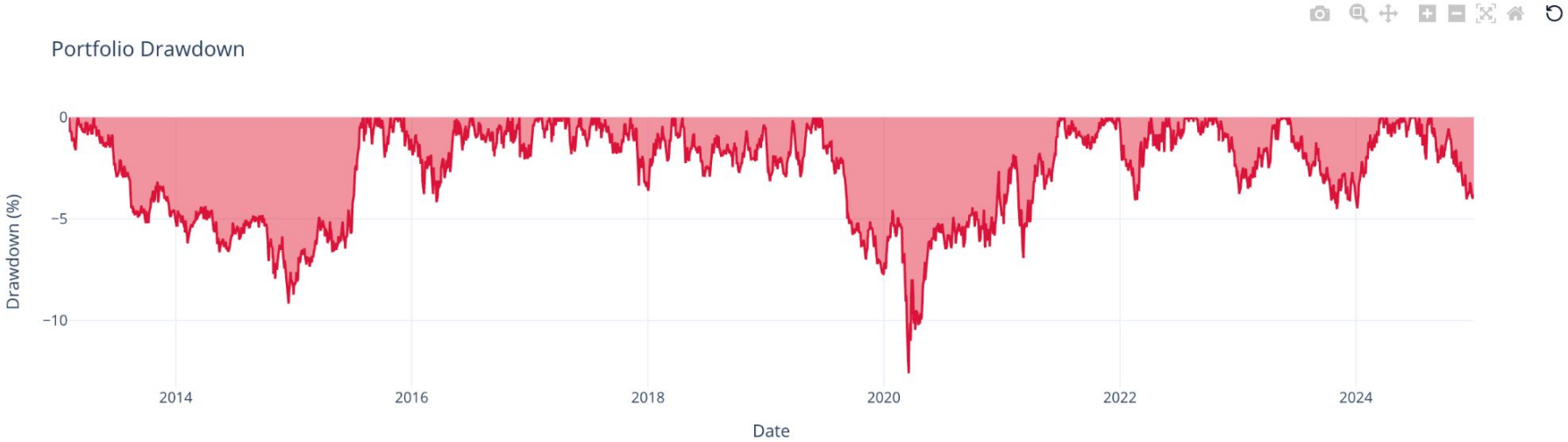
Leverage Summary

Count	Mean Leverage	Min Leverage	Max Leverage	Std Leverage
2998	401.2	265.6	532.71	63.05

Cumulative Returns



Drawdown


Portfolio Drawdown
0
−5
−10
Drawdown (%)
2014
2016
2018
2020
2022
2024
Date

Idiosyncratic Momentum

Date range: 2013-02-04 → 2024-12-31 **(2998 trading days)**

Performance Summary

Count	Mean Return (%)	Volatility (%)	Total Return (%)	Sharpe
2998	2.52	5	33	0.5

Drawdown Summary

Count	Mean Drawdown (%)	Max Drawdown (%)	Current Drawdown (%)	Longest Drawdown (days)
2998	-2.78	-8.55	-1.94	576

Leverage Summary

Count	Mean Leverage	Min Leverage	Max Leverage	Std Leverage
2998	449.43	320.68	559.71	65.3

Cumulative Returns

Drawdown

Portfolio Drawdown
0
−2
−4
−6
−8
Drawdown (%)
2014
2016
2018
2020
2022
2024
Date

Correlation Table of Portfolio Returns

standard_vs_industry --- f64	standard_vs_idiosyncratic --- f64	industry_vs_idiosyncratic --- f64
0.253964	0.790854	0.1322

Thesis / Intuition

- Standard (individual stock) momentum is a well-known anomaly where stocks that did well in the past 6–12 months tend to keep doing well in the near future, and past losers keep losing.
- Industry-level news (regulatory shifts, commodity cycles, tech waves) diffuses slowly across investors. Prices don't fully adjust immediately, so the whole sector drifts in the direction of the news for months. Because stocks within an industry move together, this drift gets picked up by standard momentum strategies, making the strategy look diversified when it actually isn't.
- Our Strategy: Invest in stocks whose industry is outperforming and short stocks whose industry is underperforming

Next Steps

- Constrain backtester to mimic investment in the industry factors themselves. We will do this with ratio constraints:
  - Take a cadidate industry i with m_i stocks.
  - Number the stocks in industry 1…m_j
  - Create a (m_i - 1) x N matrix